HACKATHON PROJECT · LAB AUTOMATION

Roboswap

A tool changer for benchtop liquid handling

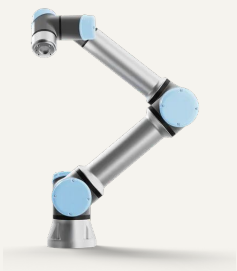


Philip, Neil, Apurva, Maya, Leo

THE PROBLEM

Lab automation, mostly boxes and arms, why do we need a box?



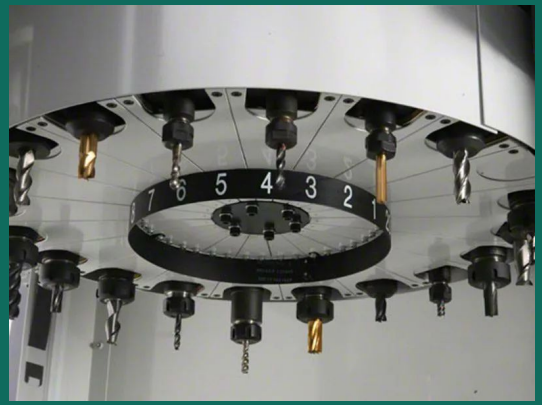






THE IDEA

Tool changers are useful





Swap the tool,
not the robot.

THE IDEA

Make a tool changer for lab automation

Robot arm can swap:

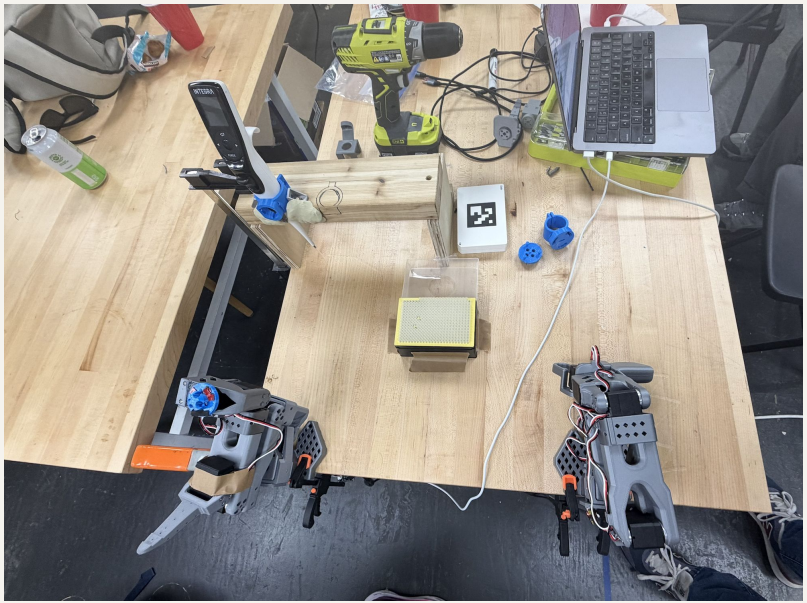
- Single channel pipette
- Multi channel pipette
- 96/384 head
- Grippers, decappers



WHAT WE BUILT

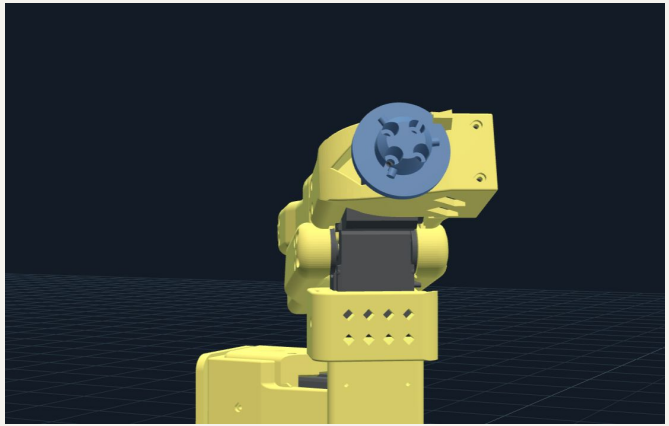
Roboswap: Robot arm with a swappable pipette

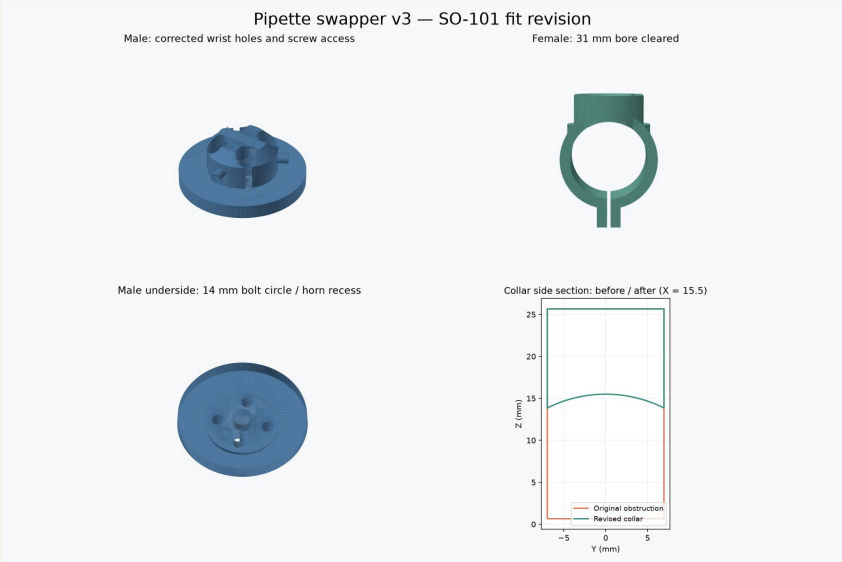
- SO-101 arm with its stock jaw gripper
- Integra electronic pipette, 34 mm body
- 3D-printed collar that the jaws lock onto
- Wood-block holder with a 33 mm hole
- Wrist camera for labware

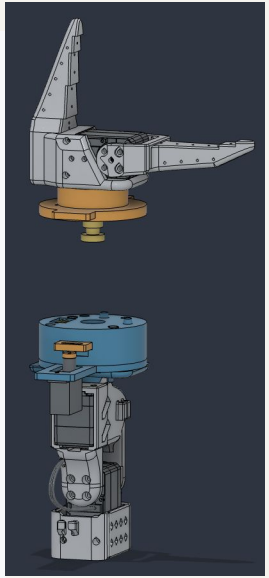


WHAT WE BUILT

Physical hardware

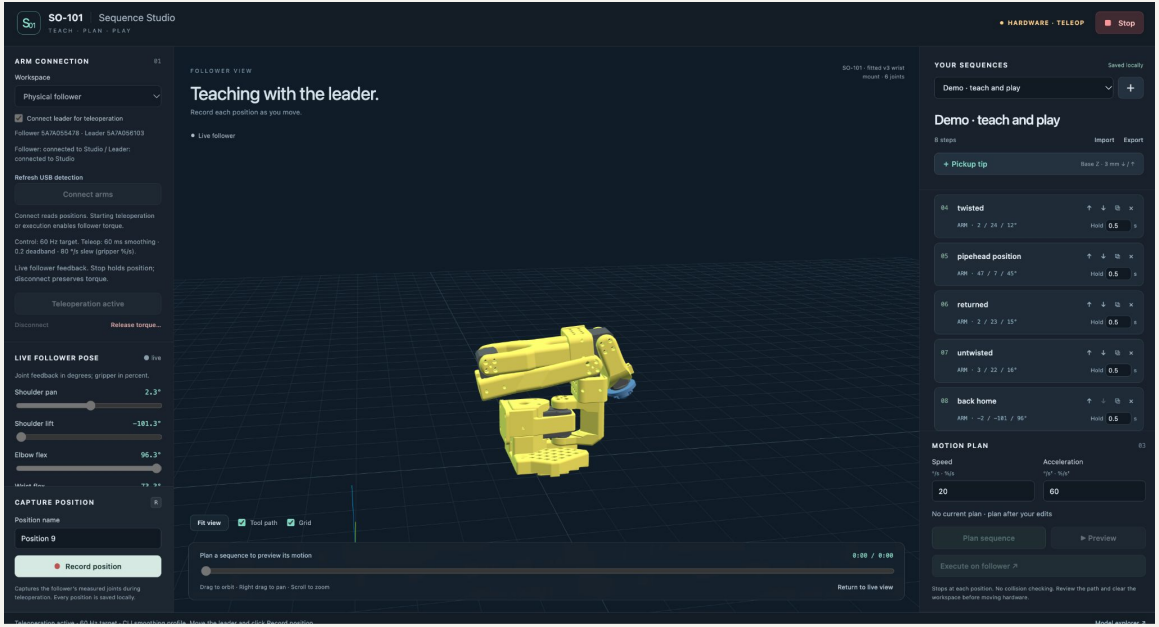






WHAT WE BUILT

Simulation, save movement sequences

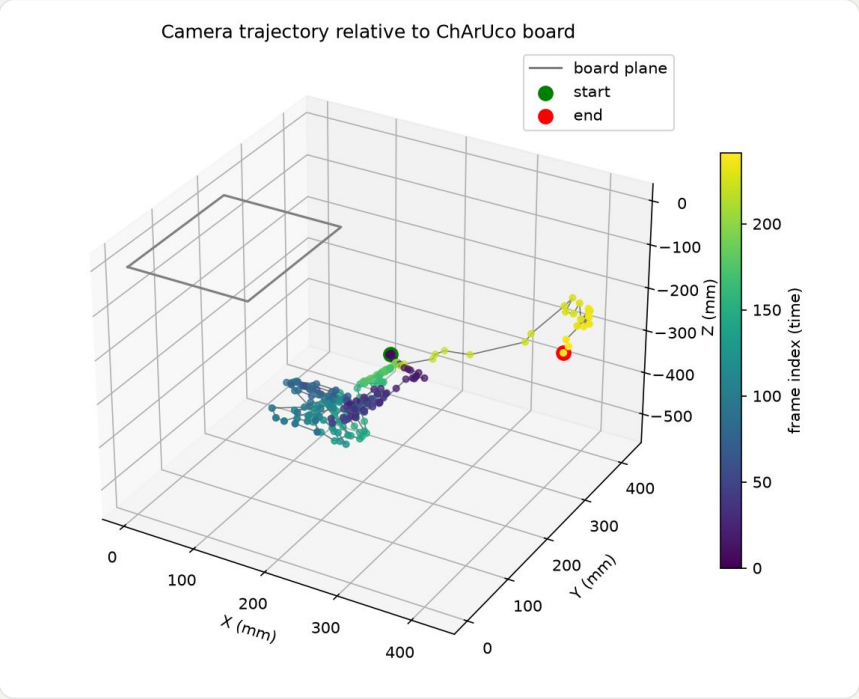


WHAT WE BUILT

Vision system

- **Calibration** — ChArUco board removes lens distortion
- **Pose tracking** — camera's 6-DOF motion vs. the board, shown right
- **Tip detection** — pipette tips registered into board coordinates

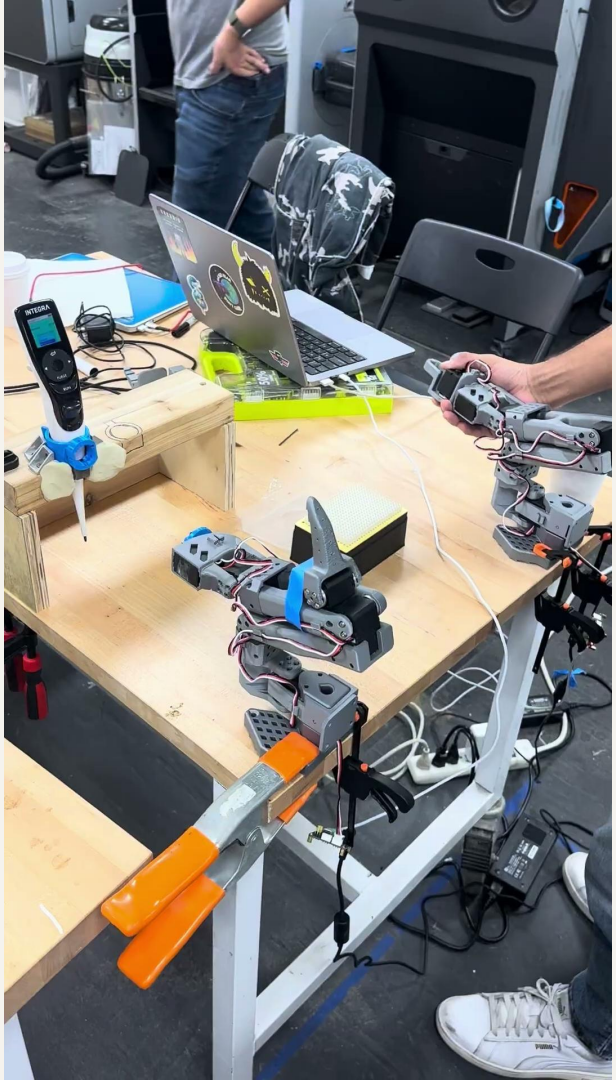
Same pose math is designed to register the SO-101 arm next — enabling motion verification and precise tool-change docking.



ChArUco board · OpenCV · SO-101 (LeRobot)

WHAT WE BUILT

Demo Video



What worked, what failed

Worked

- Simulation sequencing
- Teleoperated attachment

Fails

- Locking mechanism relied on friction, finicky to 3d print correctly
- Automated attachment, s0-101 was not precise
- Vision tracking, did not have time to integrate camera

Future goals

Build out a robust tool locking mechanism

Use high precision robot arm

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Thank you

Thanks Mark for helping with 3d prints!

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